

Solution Manual

Analysis and Design of Elementary MOS Amplifier Stages

**- First Edition -
Chapter 6**

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restart : with(Units[Standard]) : with(plots) : interface(imaginaryunit=j) : read "Parameters.mpl" :

6.1

$$g_m := 1 \text{ [mS]} : r_o := 100 \text{ [k}\Omega\text{]} :$$

$$\textbf{(a)} \quad R_L := 100 \text{ [}\Omega\text{]} :$$

First note that RS is irrelevant. The open circuit voltage gain and output resistance of the CS-CS cascade are given by:

$$A_{vl} := (g_m \cdot r_o)^2 = 10000$$

$$R_{out1} := r_o :$$

The overall voltage gain is:

$$A_{vl}' := 1 \cdot A_{vl} \cdot \frac{R_L}{R_L + R_{out1}} = 9.990$$

In essence, the gain of the CS-CS amplifier is "destroyed" by the loading of RL. For the CS-CD amplifier, we have:

$$A_{v2} := g_m \cdot r_o \cdot \frac{g_m \cdot r_o}{1 + g_m \cdot r_o} = 99.010$$

$$R_{out2} := \frac{1}{g_m + \frac{1}{r_o}} = 990.099 \text{ [}\Omega\text{]}$$

The overall voltage gain is:

$$A_{v2}' := 1 \cdot A_{v2} \cdot \frac{R_L}{R_L + R_{out2}} = 9.083$$

The open-circuit voltage gain of the CS-CD amplifier is much smaller than that of the CS-CS circuit, but it achieves almost the same overall voltage gain, since its output resistance is much smaller.

$$\textbf{(b)} \quad R_L := 10 \text{ [k}\Omega\text{]} :$$

The overall voltage gain is of the CS-CS amplifier is now:

$$A_{vl}' := 1 \cdot A_{vl} \cdot \frac{R_L}{R_L + R_{out1}} = 909.091$$

With less loading, the CS stage now achieves a much higher gain. The overall voltage gain of the CS-CD stage is now:

$$A_{v2}' := 1. A_{v2} \cdot \frac{R_L}{R_L + R_{out2}} = 90.090$$

The voltage gain of the CS-CD circuit also improves, but is now limited by the low open-circuit voltage gain.

$$(c) \ g_m := 100 \llcorner uS \llcorner : r_o := 10 \llcorner M\Omega \llcorner : R_L := 100 \llcorner \Omega \llcorner :$$

The amplifier parameters are now:

$$A_{v1} := (g_m \cdot r_o)^2 = 1000000$$

$$R_{out1} := r_o :$$

$$A_{v2} := g_m \cdot r_o \cdot \frac{g_m \cdot r_o}{1. + g_m \cdot r_o} = 999.001$$

$$R_{out2} := \frac{1.}{g_m + \frac{1}{r_o}} = 9990.010 \llcorner \Omega \llcorner$$

The overall gain values for $R_L = 100\Omega$ are:

$$A_{v1}' := 1. A_{v1} \cdot \frac{R_L}{R_L + R_{out1}} = 10.000$$

$$A_{v2}' := 1. A_{v2} \cdot \frac{R_L}{R_L + R_{out2}} = 9.901$$

The overall gain values for $R_L = 10k\Omega$ are:

$$R_L := 10 \llcorner k\Omega \llcorner :$$

$$A_{v1}' := 1. A_{v1} \cdot \frac{R_L}{R_L + R_{out1}} = 999.001$$

$$A_{v2}' := 1. A_{v2} \cdot \frac{R_L}{R_L + R_{out2}} = 499.750$$

We see again that the CS-CS circuit outperforms the CS-CD circuit when the load resistance is large.

6.2

$$g_m := 1 \text{ [mS]} : r_o := 100 \text{ [k}\Omega\text{]} :$$

$$\text{(a)} \quad R_L := 100 \text{ [}\Omega\text{]} :$$

First note that RS is irrelevant. The short circuit transconductance and output resistance of the CS-CS cascade are given by:

$$G_{m1} := g_m \cdot r_o \cdot g_m = 100 \text{ [mS]}$$

$$R_{out1} := r_o :$$

The overall transconductance is:

$$G_{m1}' := 1 \cdot G_{m1} \cdot \frac{R_{out1}}{R_L + R_{out1}} = 99.900 \text{ [mS]}$$

The overall transconductance is very close to the short circuit transconductance because the load resistance is small (close to a short). For the CS-CG amplifier, we have (approximately):

$$G_{m2} := 1 \cdot g_m = 1.000 \text{ [mS]}$$

$$R_{out2} := g_m \cdot r_o^2 = 10000000 \text{ [}\Omega\text{]} \xrightarrow{\text{replace units}} 10 \text{ [M}\Omega\text{]}$$

The overall transconductance is:

$$G_{m2}' := 1 \cdot G_{m2} \cdot \frac{R_{out2}}{R_L + R_{out2}} = 1.000 \text{ [mS]}$$

The short-circuit transconductance of the CS-CG amplifier is limited to gm; this is why the overall transconductance is much lower than that of the CS-CS circuit.

$$\text{(b)} \quad R_L := 10 \text{ [k}\Omega\text{]} :$$

The overall transconductance of the CS-CS circuit is now:

$$G_{m1}' := 1 \cdot G_{m1} \cdot \frac{R_{out1}}{R_L + R_{out1}} = 90.909 \text{ [mS]}$$

The change compared to part (a) is small, since RL is still significantly larger than Rout1. The overall transconductance of the CS-CG circuit is

$$G_{m2}' := 1 \cdot G_{m2} \cdot \frac{R_{out2}}{R_L + R_{out2}} = 0.999 \text{ [mS]}$$

Compared to (a) both numbers reduce slightly due to the increased RL.

$$(c) \ g_m := 100 \llbracket uS \rrbracket : r_o := 10 \llbracket M\Omega \rrbracket : R_L := 100 \llbracket \Omega \rrbracket :$$

The amplifier parameters are now:

$$G_{m1} := g_m \cdot r_o \cdot g_m = 100000 \llbracket uS \rrbracket \xrightarrow{\text{replace units}} 100 \llbracket mS \rrbracket$$

$$R_{out1} := r_o :$$

$$G_{m2} := 1 \cdot g_m = 100.000 \llbracket uS \rrbracket$$

$$R_{out2} := g_m \cdot r_o^2 = 10000000000 \llbracket \Omega \rrbracket \xrightarrow{\text{replace units}} 10000 \llbracket M\Omega \rrbracket$$

The overall gain values for $R_L = 100\Omega$ are:

$$G_{m1}' := 1 \cdot G_{m1} \cdot \frac{R_{out1}}{R_L + R_{out1}} = 99999.000 \llbracket uS \rrbracket \xrightarrow{\text{replace units}} 99.999 \llbracket mS \rrbracket$$

$$G_{m2}' := 1 \cdot G_{m2} \cdot \frac{R_{out2}}{R_L + R_{out2}} = 100.000 \llbracket uS \rrbracket$$

The overall gain values for $R_L = 10k\Omega$ are:

$$R_L := 10 \llbracket k\Omega \rrbracket :$$

$$G_{m1}' := 1 \cdot G_{m1} \cdot \frac{R_{out1}}{R_L + R_{out1}} = 99900.100 \llbracket uS \rrbracket \xrightarrow{\text{replace units}} 99.900 \llbracket mS \rrbracket$$

$$G_{m2}' := 1 \cdot G_{m2} \cdot \frac{R_{out2}}{R_L + R_{out2}} = 100.000 \llbracket uS \rrbracket$$

We see that the situation has not changed much compared to (a) and (b). Since R_L is in both cases much smaller than R_{out} , the CS-CS circuit wins. The CG stage helps improve the current division at the output, but this is numerically insignificant.

6.3

$$I_B := 200 \text{ [uA]} : V_{DD} := 2.5 \text{ [V]} : V_{SS} := -2.5 \text{ [V]} : W := 50 \text{ [um]} : L := 1 \text{ [um]} :$$

(a) We start by computing the voltage at the source of M1. The backgate effect must be considered for M1 and this will involve iterations. In the first iteration we ignore the backgate effect, and then recompute the threshold voltage until the result converges.

$$V_{OVI} := \sqrt{\frac{2 \cdot I_B}{\mu_n C_{ox} \cdot \frac{W}{L}}} = 0.400 \text{ [V]}$$

$$V_{SI} := -V_{T0n} - V_{OVI} = -0.900 \text{ [V]}$$

Second iteration:

$$V_{SBI} := V_{SI} - V_{SS} = 1.600 \text{ [V]}$$

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{SBI}} - \sqrt{2 \cdot \phi_f} \right) = 0.893 \text{ [V]}$$

$$V_{SI} := -V_{Tn} - V_{OVI} = -1.293 \text{ [V]}$$

Third iteration:

$$V_{SBI} := V_{SI} - V_{SS} = 1.207 \text{ [V]}$$

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{SBI}} - \sqrt{2 \cdot \phi_f} \right) = 0.813 \text{ [V]}$$

$$V_{SI} := -V_{Tn} - V_{OVI} = -1.213 \text{ [V]}$$

Fourth iteration:

$$V_{SBI} := V_{SI} - V_{SS} = 1.287 \text{ [V]}$$

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{SBI}} - \sqrt{2 \cdot \phi_f} \right) = 0.830 \text{ [V]}$$

$$V_{SI} := -V_{Tn} - V_{OVI} = -1.230 \text{ [V]}$$

Fifth iteration:

$$V_{SBI} := V_{SI} - V_{SS} = 1.270 \text{ [V]}$$

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{SBI}} - \sqrt{2 \cdot \phi_f} \right) = 0.827 \text{ [V]}$$

$$V_{SI} := -V_{Tn} - V_{OVI} = -1.227 \text{ [V]}$$

$$V_{SBI} := V_{SI} - V_{SS} = 1.273 \text{ [V]}$$

We now compute VOUT:

$$V_{OV2} := \sqrt{\frac{2 \cdot I_B}{\mu_p C_{ox} \cdot \frac{W}{L}}} = 0.566 \text{ [V]}$$

$$V_{OUT} := V_{SI} + |V_{T0p}| + V_{OV2} = -0.161 \text{ [V]}$$

(b) The open-circuit voltage gain of the second stage is unity (since we neglect r_o). The voltage gain of the first stage is reduced by g_{mb1} .

$$A_{v1} = \frac{1}{1 + \frac{g_{mb1}}{g_{m1}}} :$$

$$A_{v1} := \frac{1}{1 + \frac{\gamma_n}{2\sqrt{2 \cdot \phi_f + V_{SB1}}}} = 0.828$$

$$A_{v2} := 1 :$$

$$A_v := A_{v1} \cdot A_{v2} = 0.828$$

(c) To answer this question, we first need to compute the output resistance of the amplifier, which is simply equal to $1/g_{m2}$.

$$g_{m2} := \frac{2 \cdot I_B}{V_{OV2}} = 0.001 \text{ [S]} \xrightarrow{\text{replace units}} 0.707 \text{ [mS]}$$

$$R_{out} := \frac{1}{g_{m2}} = 1414.214 \text{ [\Omega]}$$

The overall voltage gain is:

$$A_{vprime} = A_v \cdot \frac{R_L}{R_L + R_{out}} :$$

We can solve this expression for R_L to find the desired bound:

$$A_{vprime} := 0.6 :$$

$$R_L := \frac{R_{out}}{\frac{A_v}{A_{vprime}} - 1} = 3728.484 \text{ [\Omega]}$$

The load resistor must be larger than this value to maintain a voltage gain greater than 0.6.

6.4

$$I_B := 200 \text{ } \llbracket \mu A \rrbracket : V_{DD} := 2.5 \text{ } \llbracket V \rrbracket : V_{SS} := -2.5 \text{ } \llbracket V \rrbracket : W := 50 \text{ } \llbracket \mu m \rrbracket : L := 1 \text{ } \llbracket \mu m \rrbracket :$$

(a) We start by computing the voltage at the source of M1.

$$V_{OV1} := \sqrt{\frac{2 \cdot I_B}{\mu_n C_{ox} \cdot \frac{W}{L}}} = 0.400 \text{ } \llbracket V \rrbracket$$

$$V_{S1} := -V_{T0n} - V_{OV1} = -0.900 \text{ } \llbracket V \rrbracket$$

To compute the output voltage, we need to take the backgate effect of M2 into account. This requires iterations, as in problem 6.3.

$$V_{OV2} := \sqrt{\frac{2 \cdot I_B}{\mu_p C_{ox} \cdot \frac{W}{L}}} = 0.566 \text{ } \llbracket V \rrbracket$$

$$V_{OUT} := V_{S1} + |V_{T0p}| + V_{OV2} = 0.166 \text{ } \llbracket V \rrbracket$$

Second iteration:

$$V_{BS1} := V_{DD} - V_{OUT} = 2.334 \text{ } \llbracket V \rrbracket$$

$$V_{Tp} := V_{T0p} - \gamma_p \cdot \left(\sqrt{2 \cdot \phi_f + V_{BS1}} - \sqrt{2 \cdot \phi_f} \right) = -1.026 \text{ } \llbracket V \rrbracket$$

$$V_{OUT} := V_{S1} + |V_{Tp}| + V_{OV2} = 0.691 \text{ } \llbracket V \rrbracket$$

Third iteration:

$$V_{BS1} := V_{DD} - V_{OUT} = 1.809 \text{ } \llbracket V \rrbracket$$

$$V_{Tp} := V_{T0p} - \gamma_p \cdot \left(\sqrt{2 \cdot \phi_f + V_{BS1}} - \sqrt{2 \cdot \phi_f} \right) = -0.932 \text{ } \llbracket V \rrbracket$$

$$V_{OUT} := V_{S1} + |V_{Tp}| + V_{OV2} = 0.598 \text{ } \llbracket V \rrbracket$$

Fourth iteration:

$$V_{BS1} := V_{DD} - V_{OUT} = 1.902 \text{ } \llbracket V \rrbracket$$

$$V_{Tp} := V_{T0p} - \gamma_p \cdot \left(\sqrt{2 \cdot \phi_f + V_{BS1}} - \sqrt{2 \cdot \phi_f} \right) = -0.950 \text{ } \llbracket V \rrbracket$$

$$V_{OUT} := V_{S1} + |V_{Tp}| + V_{OV2} = 0.615 \text{ } \llbracket V \rrbracket$$

Fifth iteration:

$$V_{BS1} := V_{DD} - V_{OUT} = 1.885 \text{ } \llbracket V \rrbracket$$

$$V_{Tp} := V_{T0p} - \gamma_p \cdot \left(\sqrt{2 \cdot \phi_f + V_{BS1}} - \sqrt{2 \cdot \phi_f} \right) = -0.946 \text{ } \llbracket V \rrbracket$$

$$V_{OUT} := V_{S1} + |V_{Tp}| + V_{OV2} = 0.612 \text{ } \llbracket V \rrbracket$$

(b) The open-circuit voltage gain of the second stage is unity (since we neglect r_o). The voltage gain of the first stage is reduced by g_{mb1} .

$$A_{v1} = \frac{1}{1 + \frac{g_{mb1}}{g_{m1}}} :$$

$$A_{v1} := \frac{1}{1 + \frac{\gamma_n}{2\sqrt{2 \cdot \phi_f + V_{SB1}}}} = 0.828$$

$$A_{v2} := 1 :$$

$$A_v := A_{v1} \cdot A_{v2} = 0.828$$

(c) To answer this question, we first need to compute the output resistance of the amplifier, which is simply equal to $1/g_{m2}$.

$$g_{m2} := \frac{2 \cdot I_B}{V_{OV2}} = 0.001 \text{ [S]} \xrightarrow{\text{replace units}} 0.707 \text{ [mS]}$$

$$R_{out} := \frac{1}{g_{m2}} = 1414.214 \text{ [\Omega]}$$

The overall voltage gain is:

$$A_{vprime} = A_v \cdot \frac{R_L}{R_L + R_{out}} :$$

We can solve this expression for R_L to find the desired bound:

$$A_{vprime} := 0.6 :$$

$$R_L := \frac{R_{out}}{\frac{A_v}{A_{vprime}} - 1} = 3728.484 \text{ [\Omega]}$$

The load resistor must be larger than this value to maintain a voltage gain greater than 0.6.

6.5

$$W_1 := 20 \llbracket \mu m \rrbracket : W_3 := 50 \llbracket \mu m \rrbracket : L := 1 \llbracket \mu m \rrbracket : V_{SS} := -3 \llbracket V \rrbracket : V_{DD} := 3 \llbracket V \rrbracket : I_D := 500 \llbracket \mu A \rrbracket : R_B := 6 \llbracket k\Omega \rrbracket : R_L := 1 \llbracket k\Omega \rrbracket :$$

(a) We first compute the source-gate voltage of M2B, which then allows us to compute W2B.

$$V_{SG2B} := V_{DD} - (V_{SS} + I_D \cdot R_B) = 3 \llbracket V \rrbracket$$

$$W_{2B} := \frac{I_D}{\frac{1}{2} \mu_p C_{ox} \cdot \frac{1}{L} \cdot (V_{SG2B} + V_{T0p})^2} = 0.000 \llbracket m \rrbracket \xrightarrow{\text{replace units}} 6.400 \llbracket \mu m \rrbracket$$

(b)

$$V_{OV3} := \sqrt{\frac{2 \cdot I_D}{\frac{1}{2} \mu_p C_{ox} \cdot \frac{W_3}{L}}} = 1.265 \llbracket V \rrbracket$$

$$V_{G3} := 0 \llbracket V \rrbracket - V_{OV3} - |V_{T0p}| = -1.765 \llbracket V \rrbracket$$

(c)

$$V_{OV1} := \sqrt{\frac{2 \cdot I_D}{\frac{1}{2} \mu_n C_{ox} \cdot \frac{W_1}{L}}} = 1.414 \llbracket V \rrbracket$$

$$V_{BIAS} := V_{T0n} + V_{OV1} = 1.914 \llbracket V \rrbracket$$

(d) It makes sense to model this circuit using a voltage amplifier two-port. The input resistance is infinite and the remaining component values are computed as follows:

$$R_{out} := \frac{V_{OV3}}{2 \cdot I_D} = 1264.911 \llbracket \Omega \rrbracket$$

$$\lambda_1 := \frac{\lambda_{Ln}}{L} = 0.100 \llbracket \frac{1}{V} \rrbracket$$

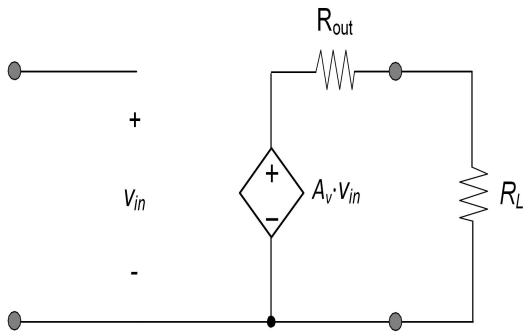
$$\lambda_2 := \frac{\lambda_{Lp}}{L} = 0.100 \llbracket \frac{1}{V} \rrbracket$$

$$r_{o1} := \frac{1 + \lambda_1 \cdot (V_{G3} - V_{SS})}{\lambda_1 \cdot I_D} = 22470.178 \llbracket \Omega \rrbracket \xrightarrow{\text{replace units}} 22.470 \llbracket k\Omega \rrbracket$$

$$r_{o2} := \frac{1 + \lambda_2 \cdot (V_{DD} - V_{G3})}{\lambda_2 \cdot I_D} = 29529.822 \llbracket \Omega \rrbracket \xrightarrow{\text{replace units}} 29.530 \llbracket k\Omega \rrbracket$$

$$g_{m1} := \frac{2 \cdot I_D}{V_{OV1}} = 0.001 \llbracket S \rrbracket \xrightarrow{\text{replace units}} 0.707 \llbracket mS \rrbracket$$

$$A_v := g_{m1} \cdot \frac{r_{o1} \cdot r_{o2}}{r_{o1} + r_{o2}} \cdot 1 = 9.023$$



(e)

$$A_{vprime} := A_v \cdot \frac{R_L}{R_L + R_{out}} = 3.984$$

6.6

$$W := 40 \llbracket \mu m \rrbracket : L := 1.5 \llbracket \mu m \rrbracket : I_{REF} := 100 \llbracket \mu A \rrbracket : G_m := 1 \llbracket mS \rrbracket : R_L := 1 \llbracket k\Omega \rrbracket : R_S := 10 \llbracket k\Omega \rrbracket : \\ V_{DD} := 5 \llbracket V \rrbracket : V_{SS} := -5 \llbracket V \rrbracket :$$

(a) Approximating the current gain of M2 as unity, G_m is simply equal to g_{m1} .

$$g_{m1} := G_m : \\ V_{OV1} := \frac{2 \cdot I_{REF}}{g_{m1}} = 0.200 \llbracket V \rrbracket \\ W_1 := \frac{I_{REF}}{\frac{1}{2} \mu n C_{ox} \cdot \frac{1}{L} \cdot V_{OV1}^2} = 0.000 \llbracket m \rrbracket \xrightarrow{\text{replace units}} 150.000 \llbracket \mu m \rrbracket$$

(b)

$$V_{BIAS} := V_{T0n} + V_{OV1} = 0.700 \llbracket V \rrbracket$$

(c) For simplicity, we ignore the $\lambda \cdot V_{DS}$ terms in this calculation.

$$\lambda := \frac{\lambda_{Ln}}{L} = 0.067 \llbracket \frac{1}{V} \rrbracket \\ r_o := \frac{1}{\lambda \cdot I_{REF}} = 1.500 \cdot 10^5 \llbracket \Omega \rrbracket \xrightarrow{\text{replace units}} 150.000 \llbracket k\Omega \rrbracket \\ g_{m3} := \sqrt{2 \cdot I_{REF} \cdot \mu p C_{ox} \cdot \frac{W}{L}} = 0.000 \llbracket S \rrbracket \xrightarrow{\text{replace units}} 0.365 \llbracket mS \rrbracket \\ g_{m2} := \sqrt{2 \cdot I_{REF} \cdot \mu n C_{ox} \cdot \frac{W}{L}} = 0.001 \llbracket S \rrbracket \xrightarrow{\text{replace units}} 0.516 \llbracket mS \rrbracket \\ R_{out} := \left(\frac{1}{r_o \cdot (1 + g_{m2} \cdot r_o)} + \frac{1}{r_o \cdot (1 + g_{m3} \cdot r_o)} \right)^{-1} = 4.890 \cdot 10^6 \llbracket \frac{1}{S} \rrbracket \xrightarrow{\text{replace units}} 4.890 \llbracket M\Omega \rrbracket$$

(d)

$$G_{mprime} := G_m \cdot \frac{R_{out}}{R_{out} + R_L} = .9997955 \llbracket mS \rrbracket$$

(e) The required calculations are similar to Example 6-4. We consider only three capacitors: C_{gs1} , C_{gd1} , and C_L , which is comprised of the junction capacitances of M2 and M3. The time constants due to the capacitors at the drain node of M1 are typically negligible, as observed in Example 6-4.

$$C_{gs1} := \frac{2}{3} \cdot W_1 \cdot L \cdot C_{ox} = 3.450 \cdot 10^{-13} \llbracket F \rrbracket \xrightarrow{\text{replace units}} 345.000 \llbracket fF \rrbracket \\ C_{gd1} := W_1 \cdot C_{ov} = 7.500 \cdot 10^{-14} \llbracket F \rrbracket \xrightarrow{\text{replace units}} 75.000 \llbracket fF \rrbracket \\ AS := 1 \cdot W \cdot L_{diff} = 1.200 \cdot 10^{-10} \llbracket m^2 \rrbracket \xrightarrow{\text{replace units}} 120.000 \llbracket \mu m^2 \rrbracket$$

$$PS := 2 \cdot L_{diff} + W = 46 \text{ } \llbracket um \rrbracket$$

$$C_{db2} := \frac{C_{Jn} \cdot AS}{\left(1 + \frac{V_{DD}}{PB}\right)^{MJ}} + \frac{C_{JSWn} \cdot PS}{\left(1 + \frac{V_{DD}}{PB}\right)^{MJSW}} = 1.727 \cdot 10^{-14} \text{ } \llbracket F \rrbracket \xrightarrow{\text{replace units}} 17.273 \text{ } \llbracket fF \rrbracket$$

$$C_{db3} := \frac{C_{Jp} \cdot AS}{\left(1 + \frac{V_{DD}}{PB}\right)^{MJ}} + \frac{C_{JSWp} \cdot PS}{\left(1 + \frac{V_{DD}}{PB}\right)^{MJSW}} = 2.312 \cdot 10^{-14} \text{ } \llbracket F \rrbracket \xrightarrow{\text{replace units}} 23.119 \text{ } \llbracket fF \rrbracket$$

$$C_L := C_{db2} + C_{db3} = 4.039 \cdot 10^{-14} \text{ } \llbracket F \rrbracket \xrightarrow{\text{replace units}} 40.392 \text{ } \llbracket fF \rrbracket$$

We can now compute the associated time constants.

$$\tau_{gslo} := R_S \cdot C_{gs1} = 3.450 \cdot 10^{-9} \text{ } \llbracket s \rrbracket \xrightarrow{\text{replace units}} 3.450 \text{ } \llbracket ns \rrbracket$$

$$\tau_{gdlo} := \left(R_S + \frac{1}{g_{m2}} + R_S \cdot \frac{1}{g_{m2}} \cdot g_{m1} \right) \cdot C_{gd1} = 2.348 \cdot 10^{-9} \text{ } \llbracket s \rrbracket \xrightarrow{\text{replace units}} 2.348 \text{ } \llbracket ns \rrbracket$$

$$\tau_{CLo} := R_L \cdot C_L = 4.039 \cdot 10^{-11} \text{ } \llbracket s \rrbracket \xrightarrow{\text{replace units}} 0.040 \text{ } \llbracket ns \rrbracket$$

The bandwidth estimate is therefore:

$$f_{3dB} := \frac{1}{2 \cdot 3.142} \cdot \frac{1}{\tau_{gslo} + \tau_{gdlo} + \tau_{CLo}} = 2.726 \cdot 10^7 \text{ } \llbracket \frac{1}{s} \rrbracket \xrightarrow{\text{replace units}} 27.258 \text{ } \llbracket MHz \rrbracket$$

6.7

$$g_m := 'g_m'; r_o := 'r_o'; C_L := 'C_L';$$

(a) To find Z_x , we apply a test voltage source at the source node of M1 and write KCL:

$$Eq1 := 0 = g_m \cdot v_t + \frac{v_t - v_{out}}{r_o} - i_t;$$

$$Eq2 := 0 = -g_m \cdot v_t + s \cdot C_L \cdot v_{out} + \frac{v_{out} - v_t}{r_o};$$

$$\text{solve}(\{Eq1, Eq2\}, [v_{out}, v_t])$$

$$\left[\left[v_{out} = \frac{i_t}{s C_L}, v_t = \frac{i_t (s C_L r_o + 1)}{s C_L (g_m r_o + 1)} \right] \right] \quad (1)$$

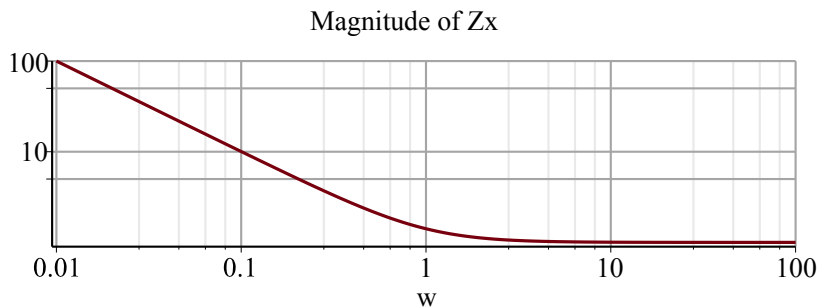
Assuming $g_m r_o \gg 1$ we have:

$$Z_x = \frac{v_t}{i_t} = \frac{1}{g_m} \frac{1 + s \cdot r_o \cdot C_L}{s \cdot r_o \cdot C_L};$$

(b) The plot below is for a normalized version of the above expression. The high frequency asymptote corresponds to $1/g_m$. The corner frequency corresponds to $1/(r_o C_L)$.

$$H := s \rightarrow \frac{1 + s}{s};$$

$$\text{loglogplot}(|H(j \cdot w)|, w = 10^{-2} .. 10^2, \text{gridlines}, \text{title} = \text{"Magnitude of } Z_x", \text{labels} = ["w", ""])$$



M1 helps alleviate Miller multiplication only when Z_x is small, i.e. close to $1/g_m$ in magnitude. This is the case only for frequencies above $1/(2 \cdot \pi \cdot r_o C_L)$. The take-home is that Miller multiplication is still significant at low frequencies when the circuit is loaded only by a capacitor.

6.8

$$W_1 := 20 \llbracket um \rrbracket : L_1 := 1 \llbracket um \rrbracket : V_{BI} := 2.5 \llbracket V \rrbracket : V_X := 1 \llbracket V \rrbracket : g_{m2} := 'g_{m2}':$$

(a) We can first compute VT and IB then follows directly:

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_X} - \sqrt{2 \cdot \phi_f} \right) = 0.768 \llbracket V \rrbracket$$

$$I_B := \frac{1}{2} \mu n C_{ox} \cdot \frac{W_1}{L_1} \cdot (V_{BI} - V_X - V_{Tn})^2 = 0.000 \llbracket A \rrbracket \xrightarrow{\text{replace units}} 267.672 \llbracket uA \rrbracket$$

(b) The sum of time constants is:

$$\tau_{tot} = R_1 \cdot C_1 + R_2 \cdot C_2 :$$

We also know that:

$$R_m = R_1 \cdot g_{m2} \cdot R_2 :$$

$$\tau_{tot} = \frac{R_m}{g_{m2} \cdot R_2} \cdot C_1 + R_2 \cdot C_2 :$$

We can differentiate this expression with respect to R2 to find a minimum:

$$0 = - \frac{R_m \cdot C_1}{g_{m2}} \cdot \frac{1}{R_2^2} + C_2 :$$

$$R_2 = \sqrt{\frac{R_m \cdot C_1}{g_{m2} \cdot C_2}} :$$

Numerical values:

$$g_{m2} := 2 \llbracket mS \rrbracket : C_1 := 1 \llbracket pF \rrbracket : C_2 := 2 \llbracket pF \rrbracket : R_m := 50 \llbracket k\Omega \rrbracket :$$

$$R_2 := \sqrt{\frac{1 \cdot R_m \cdot C_1}{g_{m2} \cdot C_2}} = 3535.534 \llbracket \Omega \rrbracket \xrightarrow{\text{replace units}} 3.536 \llbracket k\Omega \rrbracket$$

$$R_1 := \frac{R_m}{g_{m2} \cdot R_2} = 7.071 \llbracket k\Omega \rrbracket$$

6.9

$$g_m := 5 \llbracket mS \rrbracket : f_T := 5 \llbracket GHz \rrbracket : R := 1 \llbracket k\Omega \rrbracket :$$

(a)

$$A_{v0} := -g_m \cdot R \cdot \frac{g_m \cdot R}{1. + g_m \cdot R} = -4.167$$

(b) There are only two time constants to consider: the Cgs of each transistor. For the CD stage, we make use of the shortcut from Example 6-2.

$$C_{gs} := \frac{g_m}{2 \cdot 3.142 \cdot f_T} = 1.591 \cdot 10^{-13} \llbracket F \rrbracket \xrightarrow{\text{replace units}} 159.134 \llbracket fF \rrbracket$$

$$\tau_1 := R \cdot C_{gs} = 1.591 \cdot 10^{-10} \llbracket s \rrbracket \xrightarrow{\text{replace units}} 159.134 \llbracket ps \rrbracket$$

$$R_{left} := R :$$

$$R_{right} := \left(\frac{1.}{R} + g_m \right)^{-1} = 166.667 \llbracket \frac{1}{S} \rrbracket \xrightarrow{\text{replace units}} 166.667 \llbracket \Omega \rrbracket$$

$$\tau_2 := (R_{left} + R_{right} - g_m \cdot R_{left} \cdot R_{right}) \cdot C_{gs} = 5.304 \cdot 10^{-11} \llbracket s \rrbracket \xrightarrow{\text{replace units}} 53.045 \llbracket ps \rrbracket$$

$$f_{3dB} := \frac{1}{2 \cdot 3.142 \cdot (\tau_1 + \tau_2)} = 7.500 \cdot 10^8 \llbracket \frac{1}{s} \rrbracket \xrightarrow{\text{replace units}} 750.000 \llbracket MHz \rrbracket$$

6.10

$$R_D := 5 \llbracket k\Omega \rrbracket : R_S := 50 \llbracket k\Omega \rrbracket : g_m := 1 \llbracket mS \rrbracket : C_{gs} := 40.7 \llbracket fF \rrbracket : C_{gd} := 10 \llbracket fF \rrbracket : C_{db} := 11.6 \llbracket fF \rrbracket : C_{sb} := 11.6 \llbracket fF \rrbracket : g_{mb2} := 0.2 \cdot g_m :$$

Note: the calculation in Example 6-4 contains a minor mistake. The calculation of τ_{xo} should use $g_m + g_{mb2}$ in the denominator. The correct calculations are repeated below. Note that the end result is changes only by an insignificant amount.

$$\tau_{gs1o} := R_S \cdot C_{gs} = 2.035 \cdot 10^{-9} \llbracket s \rrbracket$$

$$\tau_{xo} := \frac{C_{sb} + C_{db} + C_{gs}}{g_m + g_{mb2}} = 5.325 \cdot 10^{-11} \llbracket s \rrbracket$$

$$\tau_{gd1o} := \left(\frac{1}{g_m + g_{mb2}} + \left(1 + \frac{g_m}{g_m + g_{mb2}} \right) \cdot R_S \right) \cdot C_{gd} = 9.250 \cdot 10^{-10} \llbracket s \rrbracket$$

$$\tau_{d2o} := R_D \cdot (C_{gd} + C_{db}) = 1.080 \cdot 10^{-10} \llbracket s \rrbracket$$

$$f_{3dB} := \frac{1}{2 \cdot 3.142 \cdot (\tau_{gs1o} + \tau_{xo} + \tau_{gd1o} + \tau_{d2o})} = 5.098 \cdot 10^7 \llbracket \frac{1}{s} \rrbracket \xrightarrow{\text{replace units}} 50.984 \llbracket MHz \rrbracket$$

(a) Doubling the width of M2 will increase g_{m2} by a factor of root 2 (IB is constant). C_{gs} , C_{sb} and C_{db} of M2 double. Three of the time constants therefore must be recomputed as follows:

$$\tau_{xo} := \frac{2 \cdot C_{sb} + C_{db} + 2 \cdot C_{gs}}{1 \cdot g_m \cdot \sqrt{2} + g_{mb2} \cdot \sqrt{2}} = 6.847 \cdot 10^{-11} \llbracket s \rrbracket$$

$$\tau_{gd1o} := \left(\frac{1}{\sqrt{2} \cdot g_m + \sqrt{2} \cdot g_{mb2}} + \left(1 + \frac{g_m}{\sqrt{2} \cdot g_m + \sqrt{2} \cdot g_{mb2}} \right) \cdot R_S \right) \cdot C_{gd} = 8.005 \cdot 10^{-10} \llbracket s \rrbracket$$

$$\tau_{d2o} := R_D \cdot (2 \cdot C_{gd} + 2 \cdot C_{db}) = 2.160 \cdot 10^{-10} \llbracket s \rrbracket$$

$$f_{3dB} := \frac{1}{2 \cdot 3.142 \cdot (\tau_{gs1o} + \tau_{xo} + \tau_{gd1o} + \tau_{d2o})} = 5.100 \cdot 10^7 \llbracket \frac{1}{s} \rrbracket \xrightarrow{\text{replace units}} 51.005 \llbracket MHz \rrbracket$$

The bandwidth increases only slightly. The time constant due to C_{gd1} is reduced, due to the reduced Miller gain, $g_{m1}/(g_{m2} + g_{mb2})$, but the effect is small since τ_{gs1} dominates.

(b) Halving the width of M2 will decrease g_{m2} by a factor of root 2 (IB is constant). C_{gs} , C_{sb} and C_{db} of M2 are cut in half.

$$\tau_{xo} := \frac{0.5 \cdot C_{sb} + C_{db} + 0.5 \cdot C_{gs}}{g_m \cdot \sqrt{0.5} + g_{mb2} \cdot \sqrt{0.5}} = 4.449 \cdot 10^{-11} \llbracket s \rrbracket$$

$$\tau_{gd1o} := \left(\frac{1}{\sqrt{0.5} \cdot g_m + \sqrt{2} \cdot g_{mb2}} + \left(1 + \frac{g_m}{\sqrt{0.5} \cdot g_m + \sqrt{0.5} \cdot g_{mb2}} \right) \cdot R_S \right) \cdot C_{gd} = 1.099 \cdot 10^{-9} \llbracket s \rrbracket$$

$$\tau_{d2o} := R_D \cdot (0.5 \cdot C_{gd} + 0.5 \cdot C_{db}) = 5.400 \cdot 10^{-11} \llbracket s \rrbracket$$

$$f_{3\,dB} := \frac{1}{2 \cdot 3.142 \cdot (\tau_{gs1o} + \tau_{xo} + \tau_{gd1o} + \tau_{d2o})} = 4.922 \cdot 10^7 \left[\frac{1}{s} \right] \xrightarrow{\text{replace units}} 49.224 \text{ [MHz]}$$

The bandwidth decreases slightly. The time constant due to Cgd1 is increased, due to the larger Miller gain, $gm1/(gm2+gmb2)$, but the effect is small since τ_{gs1} dominates.

6.11

$$R_D := 5 \llbracket k\Omega \rrbracket : R_S := 50 \llbracket k\Omega \rrbracket : g_m := 1 \llbracket mS \rrbracket : C_{gs} := 40.7 \llbracket fF \rrbracket : C_{gd} := 10 \llbracket fF \rrbracket : C_{db} := 11.6 \llbracket fF \rrbracket : C_{sb} := 11.6 \llbracket fF \rrbracket : g_{mb2} := 0.2 \cdot g_m :$$

(a) Doubling the width of M2 will increase g_{m2} by a factor of root 2 (IB is constant). C_{gs} , C_{sb} and C_{db} of M2 double. The pole frequencies are recomputed as follows:

$$f_{p1} := \frac{1}{2 \cdot 3.142} \cdot \frac{1}{R_S \cdot \left(C_{gs} + \left(1. + \frac{g_m}{\sqrt{2} \cdot g_m + \sqrt{2} \cdot g_{mb2}} \right) \cdot C_{gd} \right)} = 5.624 \cdot 10^7 \llbracket \frac{1}{s} \rrbracket \xrightarrow{\text{replace units}}$$

$$56.239 \llbracket MHz \rrbracket$$

$$f_{p2} := \frac{1}{2 \cdot 3.142} \cdot \frac{\sqrt{2} \cdot g_m + \sqrt{2} \cdot g_{mb2}}{2 \cdot C_{gs} + 2 \cdot C_{sb} + C_{db}} = 2.324 \cdot 10^9 \llbracket \frac{1}{s} \rrbracket \xrightarrow{\text{replace units}} 2.324 \llbracket GHz \rrbracket$$

$$f_{p3} := \frac{1}{2 \cdot 3.142} \cdot \frac{1}{R_D \cdot (2 \cdot C_{gd} + 2 \cdot C_{db})} = 7.367 \cdot 10^8 \llbracket \frac{1}{s} \rrbracket \xrightarrow{\text{replace units}} 0.737 \llbracket GHz \rrbracket$$

The most significant change is in f_{p3} , which drops by 50%. f_{p2} also reduces significantly from 3 GHz to 2.3 GHz. The take-home is that widening the transistor without increasing the current will adversely affect the non-dominant poles.

(b) Halving the width of M2 will decrease g_{m2} by a factor of root 2 (IB is constant). C_{gs} , C_{sb} and C_{db} of M2 are cut in half.

$$f_{p1} := \frac{1}{2 \cdot 3.142} \cdot \frac{1}{R_S \cdot \left(C_{gs} + \left(1. + \frac{g_m}{\sqrt{0.5} \cdot g_m + \sqrt{0.5} \cdot g_{mb2}} \right) \cdot C_{gd} \right)} = 5.094 \cdot 10^7 \llbracket \frac{1}{s} \rrbracket \xrightarrow{\text{replace units}}$$

$$50.935 \llbracket MHz \rrbracket$$

$$f_{p2} := \frac{1}{2 \cdot 3.142} \cdot \frac{\sqrt{0.5} \cdot g_m + \sqrt{0.5} \cdot g_{mb2}}{0.5 \cdot C_{gs} + 0.5 \cdot C_{sb} + C_{db}} = 3.577 \cdot 10^9 \llbracket \frac{1}{s} \rrbracket \xrightarrow{\text{replace units}} 3.577 \llbracket GHz \rrbracket$$

$$f_{p3} := \frac{1}{2 \cdot 3.142} \cdot \frac{1}{R_D \cdot (0.5 \cdot C_{gd} + 0.5 \cdot C_{db})} = 2.947 \cdot 10^9 \llbracket \frac{1}{s} \rrbracket \xrightarrow{\text{replace units}} 2.947 \llbracket GHz \rrbracket$$

The outcome is the opposite of part (a): narrowing the transistor without increasing the current will increase the non-dominant pole frequencies. The dominant pole frequency is somewhat reduced due increased Miller gain, though (change from 53.9 MHz to 50.9 MHz).

6.12

$$W := 40 \llbracket \mu m \rrbracket : L := 1 \llbracket \mu m \rrbracket : n := 2 : I_0 := 50 \llbracket \mu A \rrbracket : V_{DD} := 2.5 \llbracket V \rrbracket :$$

(a) We start by computing the gate overdrive voltages of all transistors.

$$V_{OV1} := \sqrt{\frac{2 \cdot I_0}{\frac{1}{2} \mu n C_{ox} \cdot \frac{W}{L}}} = 0.316 \llbracket V \rrbracket$$

$$V_{OV2} := V_{OV1} :$$

$$V_{OV3} := \sqrt{\frac{2 \cdot I_0}{\frac{1}{2} \mu n C_{ox} \cdot \frac{W}{n \cdot L}}} = 0.447 \llbracket V \rrbracket$$

$$V_O := V_{DD} - V_{T0n} - V_{OV3} = 1.553 \llbracket V \rrbracket$$

The minimum value of V2 follows from M1 operating at the edge of saturation.

$$V_{2\min} := V_{OV1} + V_{T0n} + V_{OV1} = 1.132 \llbracket V \rrbracket$$

The maximum value of V2 follows from M2 operating at the edge of saturation.

$$V_O - V_{S2} \llbracket V \rrbracket > V_{OV2} :$$

$$V_O - (V_{2\max} \llbracket V \rrbracket - V_{T0n} - V_{OV2}) > V_{OV2} :$$

$$V_{2\max} := V_O + V_{T0n} = 2.053 \llbracket V \rrbracket$$

The minimum possible value for VO follows from V2 = V2min and M2 at the edge of saturation.

$$V_{Omin} := V_{OV1} + V_{OV2} = 0.632 \llbracket V \rrbracket$$

The maximum value for VO is given by the voltage at which M3 turns off

$$V_{Omax} := V_{DD} - V_{T0n} = 2.000 \llbracket V \rrbracket$$

(b) We now consider the backgate effect. For M2, at V2=V2min, we know that VSB2=VOV1.

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{OV1}} - \sqrt{2 \cdot \phi_f} \right) = 0.597 \llbracket V \rrbracket$$

$$V_{2\min} := V_{OV1} + V_{Tn} + V_{OV1} = 1.230 \llbracket V \rrbracket$$

When V2=V2max, we have VSB2=VO-VOV2.

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + (V_O - V_{OV2})} - \sqrt{2 \cdot \phi_f} \right) = 0.820 \llbracket V \rrbracket$$

$$V_{2\max} := V_O + V_{Tn} = 2.372 \llbracket V \rrbracket$$

VOmin is unaffected by the backgate effect. Vmax, however, is affected by the backgate effect. To first order, we can estimate VTn at the Vmax value computed in part (a). Further iterations reveal the

true value.

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{Omax}} - \sqrt{2 \cdot \phi_f} \right) = 0.967 \llbracket V \rrbracket$$

$$V_{Omax} := V_{DD} - V_{Tn} = 1.533 \llbracket V \rrbracket$$

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{Omax}} - \sqrt{2 \cdot \phi_f} \right) = 0.880 \llbracket V \rrbracket$$

$$V_{Omax} := V_{DD} - V_{Tn} = 1.620 \llbracket V \rrbracket$$

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{Omax}} - \sqrt{2 \cdot \phi_f} \right) = 0.897 \llbracket V \rrbracket$$

$$V_{Omax} := V_{DD} - V_{Tn} = 1.603 \llbracket V \rrbracket$$

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{Omax}} - \sqrt{2 \cdot \phi_f} \right) = 0.893 \llbracket V \rrbracket$$

$$V_{Omax} := V_{DD} - V_{Tn} = 1.607 \llbracket V \rrbracket$$

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_{Omax}} - \sqrt{2 \cdot \phi_f} \right) = 0.894 \llbracket V \rrbracket$$

$$V_{Omax} := V_{DD} - V_{Tn} = 1.606 \llbracket V \rrbracket$$

(c)

$$A_{v0} = \frac{g_{m1}}{g_{m3} + g_{mb3} \llbracket S \rrbracket} = \frac{g_{m1}}{g_{m3}} \cdot \frac{1}{1 + \frac{g_{mb3} \llbracket S \rrbracket}{g_{m3}}} :$$

Iterate to find the exact value of VO (taking backgate effect into account)

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_O} - \sqrt{2 \cdot \phi_f} \right) = 0.884 \llbracket V \rrbracket$$

$$V_O := V_{DD} - V_{Tn} = 1.616 \llbracket V \rrbracket$$

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_O} - \sqrt{2 \cdot \phi_f} \right) = 0.896 \llbracket V \rrbracket$$

$$V_O := V_{DD} - V_{Tn} = 1.604 \llbracket V \rrbracket$$

$$V_{Tn} := V_{T0n} + \gamma_n \cdot \left(\sqrt{2 \cdot \phi_f + V_O} - \sqrt{2 \cdot \phi_f} \right) = 0.894 \llbracket V \rrbracket$$

$$V_O := V_{DD} - V_{Tn} = 1.606 \llbracket V \rrbracket$$

$$A_{v0} := \frac{\sqrt{1 \cdot n}}{1 + \frac{\gamma_n}{2\sqrt{2 \cdot \phi_f + V_O}}} = 1.185$$

6.13

$$L := 1 \text{ [um]} : I_{BIAS} := 100 \text{ [uA]} : n := 3 : W_A := 16 \text{ [um]} : W_B := 16 \text{ [um]} : W_1 := 9 \text{ [um]} : V_O := 0 \text{ [V]} : V_{DD} := 2.5 \text{ [V]} : V_{SS} := -2.5 \text{ [V]} : C_x := 100 \text{ [fF]} : C_L := 500 \text{ [fF]} : \mu n := \frac{\mu n C_{ox}}{C_{ox}} :$$

(a)

$$V_{OV1} := \sqrt{\frac{2 \cdot I_{BIAS}}{\frac{1}{2} \mu n C_{ox} \cdot \frac{W_1}{L}}} = 0.943 \text{ [V]}$$

$$V_{OV2} := V_{OV1} = 0.943 \text{ [V]}$$

$$R_1 := \frac{V_O - V_{SS}}{I_{BIAS}} = 25000.000 \text{ [\Omega]} \xrightarrow{\text{replace units}} 25.000 \text{ [kOmega]}$$

$$R_2 := \frac{R_1}{n} = 8333.333 \text{ [\Omega]} \xrightarrow{\text{replace units}} 8.333 \text{ [kOmega]}$$

(b) The results of (a) will not change with backgate effect considered. The only thing that will change is the gate voltage of M1 and M2. Similarly, if the threshold voltage changes from 0.5 V to 0.75 V, the gate voltage of M1 and M2 will rise by 250mV. The output quiescent point (VO) is unaffected.

(c)

$$V_{OVB} := \sqrt{\frac{2 \cdot I_{BIAS}}{\frac{1}{2} \mu n C_{ox} \cdot \frac{W_B}{L}}} = 0.707 \text{ [V]}$$

$$g_{mB} := \frac{2 \cdot I_{BIAS}}{V_{OVB}} = 0.000 \text{ [S]} \xrightarrow{\text{replace units}} 0.283 \text{ [mS]}$$

$$A_{v0} := g_{mB} \cdot R_1 = 7.071$$

(d) We assume $R_1 \gg 1/g_{m1}$ and $R_2 \gg 1/g_{m2}$

$$\tau_{C_{Lo}} = \frac{C_L}{g_{m2}} = \frac{C_L \cdot V_{OV2}}{2 \cdot n \cdot I_{BIAS}} :$$

$$\tau_{C_{xo}} = C_x \cdot R_1 :$$

$$\tau_{C_{gs1}} = \frac{C_{gs1}}{g_{m1}} = \left(\frac{3}{2} \cdot \frac{\mu n \cdot V_{OV1}}{L^2} \right)^{-1} :$$

$$\tau_{C_{gs2}} = C_{gs2} \cdot \left(R_1 + \frac{1}{g_{m2}} - R_1 \cdot \frac{1}{g_{m2}} \cdot g_{m2} \right) = \frac{C_{gs2}}{g_{m2}} = \tau_{C_{gs1}} :$$

$$f_{3dB} = \frac{1}{2 \cdot 3.142} \cdot \frac{1}{\frac{C_L \cdot V_{OV1}}{2 \cdot n \cdot I_{BIAS}} + C_x \cdot R_1 + \frac{4}{3} \cdot \frac{L^2}{\mu n \cdot V_{OV1}}} :$$

(e) We need to differentiate the denominator in the above expression to find the optimum.

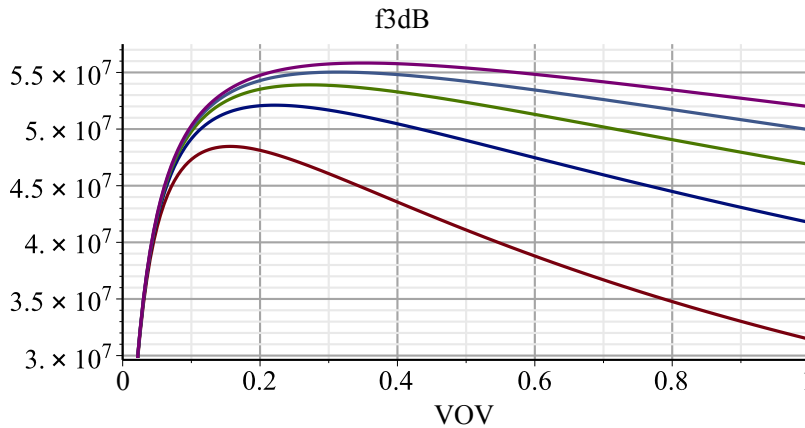
$$0 = \frac{C_L}{2 \cdot n \cdot I_{BIAS}} - \frac{4}{3} \cdot \frac{L^2}{\mu n \cdot V_{OV1}^2} :$$

$$V_{OV1} := \sqrt{\frac{4}{3} \cdot \frac{L^2}{\mu n} \cdot \frac{2 \cdot n \cdot I_{BIAS}}{C_L}} = 0.271 \text{ [V]}$$

Now verify graphically.

$$f_{3dB} := (VOV, n) \rightarrow \frac{1}{2 \cdot 3.142} \cdot \frac{1}{\frac{C_L \cdot VOV \cdot [V]}{2 \cdot n \cdot I_{BIAS}} + C_x \cdot R_1 + \frac{4}{3} \cdot \frac{L^2}{\mu n \cdot VOV \cdot [V]}} \cdot [s] :$$

`plot([seq(f3dB(VOV, n), n = [1, 2, 3, 4, 5]), VOV = 0 .. 1, gridlines, title = "f3dB", labels = ["VOV",
""])`



The maximum increases with larger n, since this helps reduce the time constant due to the large CL.

6.14

(a) The bandwidth of the cascade is found from solving the following equation:

$$\frac{1}{\sqrt{2}} = \left| \frac{1}{1 + j \cdot \frac{\omega_3 dBcasc}{\omega_3 dBstage}} \right|^N :$$

$$2^{\frac{1}{2 \cdot N}} = \left(1 + \left(\frac{\omega_3 dBcasc}{\omega_3 dBstage} \right)^2 \right)^{\frac{1}{2}} :$$

$$2^{\frac{1}{N}} - 1 = \left(\frac{\omega_3 dBcasc}{\omega_3 dBstage} \right)^2 :$$

$$\frac{\omega_3 dBcasc}{\omega_3 dBstage} = \sqrt{2^{\frac{1}{N}} - 1} :$$

(b)

$$I_{Dtot} = N \cdot I_D = N \cdot \frac{1}{2} \cdot g_m \cdot V_{OV} :$$

To eliminate VOV, we use:

$$\omega_3 dBstage = \frac{1}{R_D \cdot C_{gs}} = \frac{\omega_T}{A_{vstage}} :$$

$$\omega_T = \frac{3}{2} \cdot \frac{\mu n \cdot V_{OV}}{L^2} :$$

$$V_{OV} := \frac{2}{3} \cdot \frac{L^2}{\mu n} \cdot \omega_3 dBstage \cdot A_{vstage} :$$

Substituting gives:

$$I_{Dtot} = \frac{N}{2} \cdot g_m \cdot \frac{2}{3} \cdot \frac{L^2}{\mu n} \cdot \omega_3 dBstage \cdot A_{vstage} :$$

Next, we eliminate gm using gm = Avstage/RD:

$$I_{Dtot} = \frac{N}{2} \cdot \frac{1}{R_D} \cdot \frac{2}{3} \cdot \frac{L^2}{\mu n} \cdot \omega_3 dBstage \cdot A_{vstage}^2 :$$

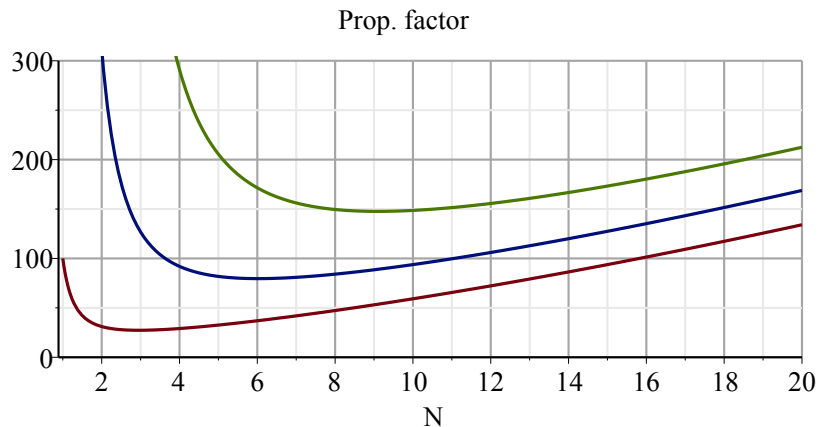
$$I_{Dtot} = \frac{1}{R_D} \cdot \frac{1}{3} \frac{L^2}{\mu n} \cdot N \frac{\omega_{3dBcasc}}{\sqrt{2^{\frac{1}{N}} - 1}} \cdot A_{vtot}^{\frac{2}{N}} :$$

The first few terms are fixed (R_D , L , μn , $\omega_{3dBcasc}$), the others are design parameters subject to optimization.

(c)

$$Propfact := (N, Avtot) \rightarrow \frac{N}{\sqrt{2^{\frac{1}{N}} - 1}} \cdot Avtot^{\frac{2}{N}} :$$

`plot([seq(Propfact(N, Avtot), Avtot = [10, 100, 1000]), N = 1 .. 20, y = 0 .. 300, gridlines, title = "Prop. factor", labels = ["N", ""])`



The optima are relatively shallow and shift to a larger number of stages for larger Avtot.

6.15

This is an open ended design problem that has no unique solution. The reader is encouraged to follow the suggested design procedure outlined in Section 6-3.